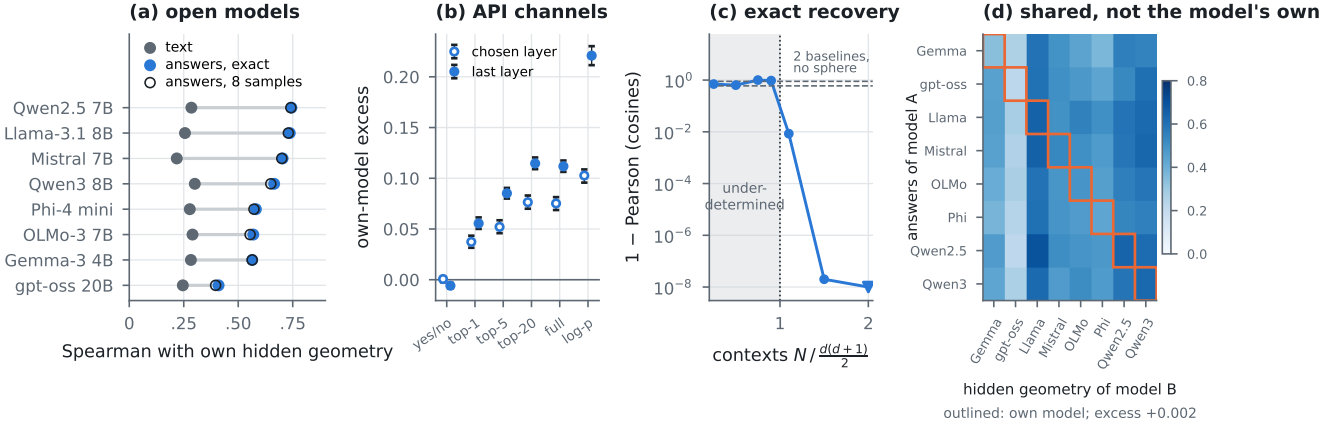
(a) open models
text
answers, exact
answers, 8 samples
Qwen2.5 7B
Llama-3.1 8B
Mistral 7B
Qwen3 8B
Phi-4 mini
OLMo-3 7B
Gemma-3 4B
gpt-oss 20B
0
.25
.50
.75
Spearman with own hidden geometry
(b) API channels
chosen layer
last layer
own-model excess
0.20
0.15
0.10
0.05
0.00
yes/no
top-1
top-5
top-20
full
log-p
(c) exact recovery
2 baselines,
no sphere
under-
determined
1 − Pearson (cosines)
10^{0}
10^{-2}
10^{-4}
10^{-6}
10^{-8}
1
2
contexts N / $\frac{d(d+1)}{2}$
(d) shared, not the model's own
answers of model A
Gemma
gpt-oss
Llama
Mistral
OLMo
Phi
Qwen2.5
Qwen3
0.8
0.6
0.4
0.2
0.0
hidden geometry of model B
outlined: own model; excess +0.002